

Java Exception Handling Lab Report

ArrayIndexOutOfBoundsException

Aim

To demonstrate exception handling in Java by catching an `ArrayIndexOutOfBoundsException` using a try-catch block.

Description

This program creates an integer array with four elements and accesses a valid index. It then attempts to access index 5, which does not exist in the array. Java throws an `ArrayIndexOutOfBoundsException`, which is caught and handled using the catch block. The program continues execution and displays a completion message.

Tools Used

- Java Development Kit (JDK)
- Java compiler (javac)
- Java Runtime Environment (JRE)
- Any Java IDE or text editor

UML / Program Structure

```
Arrayindexoutofboundexception
-----
+ main(String[] args): void
|
+--> int[] arr
+--> try block
|       |
|       +--> Access arr[5]
+--> catch(ArrayIndexOutOfBoundsException)
+--> Print successful completion message
```

Algorithm

1. Start the program.
2. Create an integer array containing 10, 20, 30, and 40.
3. Display the valid array element at index 2.
4. Enter the try block and attempt to access the element at index 5.
5. Since index 5 is outside the array bounds, an `ArrayIndexOutOfBoundsException` occurs.
6. Catch the exception and display its message.
7. Display a message indicating that the program executed smoothly.
8. Stop the program.

Complete Java Source Code

```
package exceptions;

public class Arrayindexoutofboundexception {
    public static void main(String[] args) {
        int arr[] = {10,20,30,40};
        System.out.println("validnumber" + arr[2]);
        try {
            System.out.println("invalidnumber" + arr[5]);
        } catch(ArrayIndexOutOfBoundsException e) {
            System.out.println("caught exceptions:" + e.getMessage());
        }
        System.out.println("The program is excutes smoothly");
    }
}
```

Sample Output

```
validnumber30
caught exceptions:Index 5 out of bounds for length 4
The program is excutes smoothly
```

Result

The program successfully demonstrates handling of `ArrayIndexOutOfBoundsException` using try-catch. The exception is caught, its message is displayed, and the program terminates smoothly without crashing.